FINITENESS OF A MODULE IS EQUIVALENT TO INJECTIVITY OF HOM INTO FILTERED COLIMITS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Description of a filtered colimit of modules. Used in both halves of the counted proof (lines 1818-1820 and 1827-1830). Lets a vanishing condition be moved to some stage of the colimit.
- Generation of a submodule by a finite set. Used in the converse direction (lines 1826-1830). Produces the submodule N_E whose quotient is used to test the injectivity hypothesis.

Lemma 0.1. *Let R be a ring. Let N be an R-module. The following are equivalent*

(1) *N is a finite R-module,*
(2) *for any filtered colimit $M = \operatorname{colim} M_i$ of R-modules the map $\operatorname{colim} \mathrm{Hom}_R(N, M_i) \to \mathrm{Hom}_R(N, M)$ is injective.*

Proof. Assume (1) and choose generators $x_1, \ldots, x_m$ for N. If $N \to M_i$ is a module map and the composition $N \to M_i \to M$ is zero, then because $M = \operatorname{colim}_{i' \geq i} M_{i'}$ for each $j \in \{1, \ldots, m\}$ we can find a $i' \geq i$ such that x_j maps to zero in $M_{i'}$. Since there are finitely many x_j we can find a single i' which works for all of them. Then the composition $N \to M_i \to M_{i'}$ is zero and we conclude the map is injective, i.e., part (2) holds.

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Assume (2). For a finite subset $E \subset N$ denote $N_E \subset N$ the R-submodule generated by the elements of E. Then $0 = \operatorname{colim} N/N_E$ is a filtered colimit. Hence we see that $\mathrm{id} : N \to N$ maps into N_E for some E, i.e., N is finitely generated. $\square$